

Practice — Pressure: Force per Unit Area

Section A is interactive — tap an option or type and press Check. Sections B and C are written practice.

Objective (1 mark each)

Q1 • MCQ**[1]**

Pressure is defined as —

- (a) force $\times$ area (b) force $\div$ area (c) area $\div$ force (d) force + area

Q2 • MCQ**[1]**

The SI unit of pressure is the —

- (a) newton (b) pascal (c) joule (d) metre

Q3 • MCQ**[1]**

A force of 200 N acts on 4 m². The pressure is —

- (a) 800 Pa (b) 50 Pa (c) 204 Pa (d) 0.02 Pa

Q4 • MCQ**[1]**

Bags with broad straps are comfortable because they —

- (a) increase pressure (b) reduce pressure by spreading force over a larger area
(c) reduce the weight (d) add force

Q5 • ASSERTION-REASON**[1]**

A: A sharp knife cuts more easily than a blunt one.

R: A smaller area of contact gives greater pressure for the same force.

- (a) Both true, R explains A (b) Both true, R does not explain A (c) A true, R false
(d) A false, R true

Q6 · FILL IN THE BLANK

[1]

1 pascal equals 1 newton per _____ metre.

Q7 · FILL IN THE BLANK

[1]

For the same force, a smaller area gives a _____ pressure.

Short answer (2 marks each)

Q8

[2]

Define pressure and give its SI unit.

Q9

[2]

A force of 150 N acts on an area of 5 m². Calculate the pressure.

Q10

[2]

Why do people place a round cloth pad under a load carried on the head?

Long / application (3 marks each)

Q11

[3]

Explain, using pressure, (i) why a nail is driven using its pointed end and (ii) why tractor tyres are wide.

Q12

[3]

A box of weight 240 N rests first on a face of area 6 m^2 and then on a face of area 2 m^2 . Find the pressure in each case and state which face exerts more pressure.

[Check yourself](#) → Answer key